PHYSICS ONLY

# 28 COSMOLOGY

How did the Universe begin? Did it have a beginning or did it always exist? Will it have an end or will it go on for ever? As scientists discover more and more about the Universe, they believe they are getting closer and closer to discovering the answers to these questions. The first clues to solving these problems came from observing explosions like the one shown in Figure 28.1.



▶ Figure 28.1 This photograph was taken a fraction of a second after an explosion. The movement of the particles and the energy released have some similarities to events we can see in space. Could this be a clue that suggests that the Universe started with a 'Big Bang'?

## LEARNING OBJECTIVES

- Describe the past evolution of the Universe and the main arguments in favour of the Big Bang
- Describe evidence that supports the Big Bang theory (red-shift and cosmic microwave background (CMB) radiation)
- Describe that if a wave source is moving relative to an observer there will be a change in the observed frequency and wavelength
- Use the equation relating change in wavelength, wavelength, velocity of a galaxy and the speed of light:

$$\frac{\text{change in wavelength}}{\text{reference wavelength}} = \frac{\text{velocity of a galaxy}}{\text{speed of light}}$$

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{\Delta\lambda}{\lambda_0} = \frac{v}{c}$$

- Describe the red-shift in light received from galaxies at different distances from the Earth
- Explain why the red-shift of galaxies provides evidence for the expansion of the Universe



▲ Figure 28.2 A telescope to look at stars

## THE DOPPLER EFFECT

If we look at the stars and galaxies in our Universe using a simple **optical telescope** it is impossible to tell if they are moving, and if they are moving then how fast or in which direction. If however we look at the spectrum of light emitted by a star or galaxy we can discover all of the above. We can do this by using an effect called the Doppler effect (see page 104), first described by an Austrian scientist named Christian Doppler in 1842.

Before we look at the Doppler effect in space, let us look at some more familiar situations where we can observe it.

If we close our eyes and listen to the engine of a fast moving car it is usually very simple to decide if it is approaching us or moving away. The loudness of the motor will of course change but so too will the frequency of the sound we hear. As the car approaches the frequency of the sound produced by its

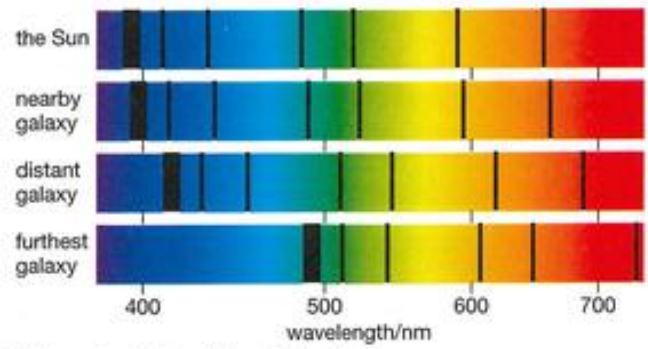



▲ Figure 28.3 An everyday example of the Doppler effect

motor appears to be high but when it moves away it appears to have a lower frequency. This apparent change in frequency is called the Doppler effect. The faster the car is moving the greater this change in frequency.

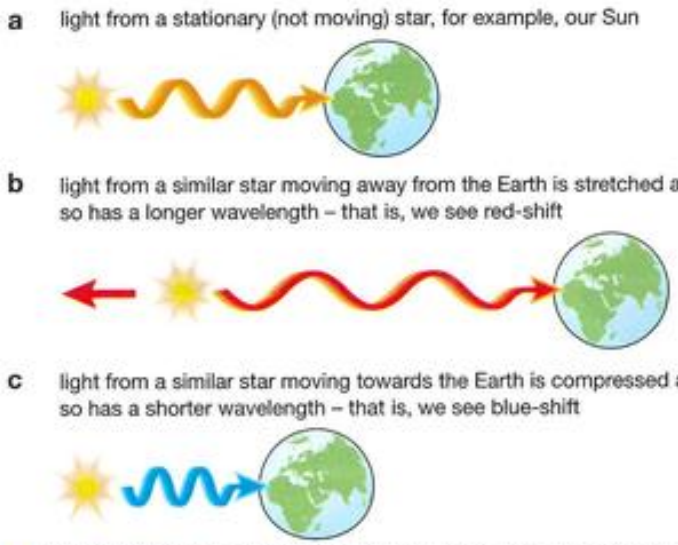
The Doppler effect is a property of all waves. It happens not only with sound waves but also with light waves. When astronomers look at light spectra from distant stars and galaxies they can see the Doppler effect.

Figure 28.4 shows the spectra from four different objects. The first is the spectrum from our Sun. The dark lines we can see are called **absorption** lines. They are the frequencies of light that have been absorbed by hydrogen. The second spectrum is from a nearby galaxy. Here we can clearly see that the positions of the dark lines have moved towards the red part of the spectrum. The third is the spectrum from a very distant galaxy – the lines have moved even further towards the red. The last spectrum shows the greatest shift for the furthest galaxy. We call this displacement '**red-shift**'. Like the changes in the frequency of the car engine it indicates a relative motion between the galaxies and the observer.



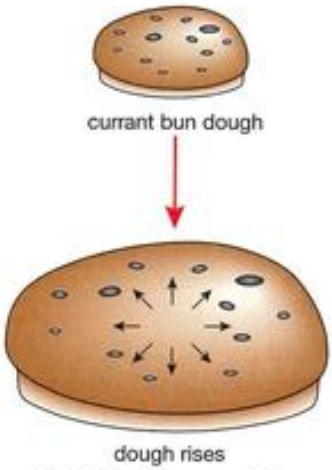

▲ Figure 28.4 Light spectra for four different objects

Red-shift indicates that the source of the light waves is moving away from the observer. Blue-shift would indicate that the source of light is moving towards the observer. We can see why this is the case in Figure 28.5.




▲ Figure 28.5 Demonstrations of light from a star moving away from Earth and moving towards it

When we compare the light emitted from all the different galaxies a clear pattern emerges. Almost all the galaxies emit light with red-shift. The further away a galaxy is the greater the red-shift and therefore the faster it is moving away from us. We see exactly this pattern in situations that involve explosions.




▲ Figure 28.6 As the bun expands the currants get further apart. Similarly as the Universe expands the galaxies get further apart.

So scientists now believe that the Universe is expanding and that at some time in the past, all the matter in the Universe was in one place just before an explosion. This theory is called the Big Bang theory.

Another way to imagine the expansion of the Universe is to observe what happens to the dough and currants in a bun when they are placed in an oven to cook. Over a period of time the dough expands and the currants all move away from each other. In this model the dough represents the Universe, increasing in size as time passes. The currants represent galaxies that are seen to be moving away from each other as the Universe expands.

## THE DOPPLER EQUATION

The equation below shows us how to calculate the speed at which a star or galaxy is moving relative to us.

$$\frac{\text{change in wavelength, } \Delta\lambda}{\text{reference wavelength, } \lambda_0} = \frac{\text{velocity of a galaxy, } v}{\text{speed of light, } c}$$

$$\frac{\lambda - \lambda_0}{\lambda_0} = \frac{\Delta\lambda}{\lambda_0} = \frac{v}{c}$$

**EXAMPLE 1**

Light emitted by the Sun has a wavelength of 434 nm. A distant galaxy emits the same light but because of the Doppler effect it has a wavelength of 482 nm. Calculate the speed at which the galaxy is moving relative to the Earth. (speed of light = $3 \times 10^8$ m/s)

$$\frac{\Delta\lambda}{\lambda_0} = \frac{v}{c}$$

$$v = \frac{\Delta\lambda}{\lambda_0} \times c$$

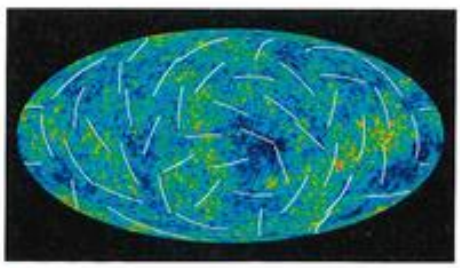
$$= \frac{(482 - 434) \times 10^{-9} \text{ m}}{434 \times 10^{-9} \text{ m}} \times 3 \times 10^8 \text{ m/s}$$

$$= \left(\frac{48}{434}\right) \times 3 \times 10^8 \text{ m/s}$$

$$= 0.332 \times 10^8 \text{ m/s}$$

$$= 33\,200 \text{ km/s}$$

Because the apparent wavelength is longer, this is an example of red-shift. The galaxy is therefore moving away from the Earth.



▲ Figure 28.7 A map of the cosmic background radiation, the afterglow of the Big Bang

## COSMIC MICROWAVE BACKGROUND (CMB) RADIATION

To reinforce the idea that the Universe began with a Big Bang, scientists searched for something else that we might connect with an explosion – that is, the release of energy. This energy would be in the form of waves. But as the Universe expands these waves would become stretched and therefore they would have much longer wavelengths than before. In fact these wavelengths were predicted to be so long that they would be in the microwave part of the electromagnetic spectrum. In the 1960s scientists detected this afterglow of energy. As was predicted, they are microwaves and can be detected in all directions in the Universe. They became known as cosmic microwave background (CMB) radiation.

## LOOKING AHEAD – DARK ENERGY

Having found evidence that the Universe began with a Big Bang and is expanding, scientists are now asking the question, 'Will the Universe continue to expand for ever or will gravity gradually slow it down and reverse the process, pulling all the matter back to a single point (the Big Crunch)?' The rate at which the expansion of the Universe slows depends upon how much mass there is in the Universe. The greater the mass the more the expansion of the Universe will be slowed.

However, in 1988, observations of a very distant galaxy made by the Hubble space telescope showed that the expansion of the Universe had previously been slower. This suggested that the expansion of the Universe is not slowing but accelerating. Scientists are still not able to explain this. They have a theory that there is energy in the Universe that is causing this acceleration. They cannot see it or detect it so it is called 'dark energy'. If you continue to study physics after your International GCSE you will learn more about this.

## CHAPTER QUESTIONS

More questions on cosmology can be found at the end of Unit 8 on page 275.

SKILLS REASONING (6)

1 A racing car travelling at high speed passes an observer. Describe the frequency of the sound produced by the car's engine heard by the observer
   a as the car approaches, and
   b as the car moves away.

SKILLS CRITICAL THINKING (5)

2 The compression of light waves emitted by a galaxy that is approaching an observer is called blue-shift. What is the stretching of light waves by a galaxy moving away from an observer called?

SKILLS REASONING (6)

3 How will the light spectra of a distant galaxy differ from that of a very distant galaxy? Explain what this difference indicates.

(5)

4 What two observed features of our Universe suggest that it began with a Big Bang?

SKILLS PROBLEM SOLVING (8)

5 The spectrum of a distant galaxy shows a change in position of absorption lines of 300 nm compared with the Sun's spectrum. If these lines in the Sun's spectrum correspond to a light with wavelength of 600 nm calculate the speed of this galaxy relative to the speed of the Earth.
(speed of light = $3 \times 10^8$ m/s)

**END OF PHYSICS ONLY**